Statement of Purpose

Niyati Bafna (PhD applicant for Fall—2023, Johns Hopkins University)

In recent years, NLP models have made tremendous progress on myriad complex natural language tasks. However, data inequity means that current neural methods only work well for a small percentage of human languages (Joshi et al., 2020). This problem requires a more nuanced and fundamental solution than simply collecting more data for each of the world's ∼7000 languages; after all, a language itself is not a monolith. Millions of people around the world speak in dialectical or code-switched *variants* of a "language"; data collection, therefore, becomes increasingly infeasible the higher the number of people NLP truly supports. Data-intensive methods, left on their own, potentially marginalize already vulnerable communities. It is our responsibility, therefore, to address the problem of multilinguality; further, it is also extremely interesting from a scientific perspective. In my opinion, the quest to make NLP models dialectically nimble, transferable, and interpretable, plumbs to the depths of what we know about linguistics and machine learning, and presents us with an immensely urgent, exciting, and challenging paradigm.

Interests: My main research interests are as follows, accompanied by examples of papers that I find inspiring in the respective field:

- Multilinguality, including problems in (a) dialectical NLP, (b) cross-lingual transfer, and (c) low-resource NLP. Can we fundamentally understand and systematize linguistic interactions with respect to how they can be exploited for cross-lingual transfer? What are the conditions under which a model learns with very little data? (Chai et al., 2022; Minixhofer et al., 2022; Abulimiti and Schultz, 2020)
- Linguistic interpretability of large language models, including questions of (a) architectural innovation, and (b) linguistic probing. What kinds of structural biases (if any) better facilitate linguistic generalization? Can we engineer tokenization, embedding dimensions, or (self-) attention structure to control what our language model is better or worse at? (Wan et al., 2020; Singh et al., 2019; Nzeyimana and Niyongabo Rubungo, 2022)
- Morphosyntax in NLP. Can we come up with general and cross-lingual answers to the debate between linguistically motivated morphological treatment and "statistical" chopping in NLP? How can we use our knowledge about shared morphological patterns, both genealogical and as regards borrowings, to aid cross-lingual transfer? (Zhu et al., 2019; Wu et al., 2021)

Thesis: I did a dual M.Sc. in Computer Science and Computational Linguistics at Charles University, Prague, and Saarland University, Germany, as an Erasmus Mundus scholar. I wrote my thesis, titled "Empirical Models for an Indic Dialect Continuum", under the co-supervision of Professors Josef van Genabith and Zdeněk Žabokrtský.[1] The project dealt in developing unsupervised strategies for cognate induction in (extremely) low-resource scenarios, data collection and exploratory cross-linguistic experiments for 26 dialects of the linguistic continuum across North India, with more than 100 million native speakers; 16 of these dialects were previously zero-resource. I implemented a novel method for unsupervised cognate induction from monolingual corpora, based on combining noisy semantic signals from bilingual spaces with orthographic cues modelling sound change, and showed that it outperformed both traditional baselines as well as custom edit distance matrices learned in an Expectation Maximisation-style algorithm. I chose this topic for two reasons: firstly, I wanted to explore the rich theoretical potential that dialect continuums hold for technical and linguistic insight, so far under-researched due to lack of data, and secondly, because I wish to contribute to language and dialect-inclusive NLP, and to building a research base for under-resourced language families.

Internships and projects: This thesis followed a previous internship project with Professor Žabokrtský, in the summer of 2021. I studied the transfer of static embeddings from a high-resource language (Hindi) to artificially low-resource genealogically related languages (Marathi, Nepali) **(Bafna and Žabokrtský, 2022)**. Starting with two monolingual embedding spaces, I employed VecMap (Artetxe et al., 2017), an EM-style algorithm that alternately optimizes the sum of cosine distances between words in a bilingual lexicon, and a bilingual lexicon given two monolingual

[1] Available here: `https://dspace.cuni.cz/handle/20.500.11956/175497`. Paper to be presented at CoNLL '22

spaces. We showed that the resulting subword embeddings are close in quality to FastText embeddings trained on two orders of magnitude more data for Marathi and Nepali. A secondary result also demonstrated via ablations the benefits of working with subwords for transfer to morphologically rich languages. This project expanded my interest in investigating multilingual models for the effects of shared genealogy and typology, with the dual aim of interpretability and data efficiency.

Two other internships that helped me hone research skills include my time at Amazon India in the summer of 2020, where I worked at improving multilingual Named Entity Recognition models that previously floundered on noun-compounding languages like German, and my work with Professor Samar Husain over 2020 in designing computational models to mimic syntactic forgetting in Hindi **(Sharma et al., 2021)**. Finally, there have been several other projects that I have used to whet my interests, and test my intuition in practice. I worked with Professor Ondřej Bojar on the problem of technical term handling in English-to-Hinglish machine translation, using synthetically created code-switching for training **(Bafna et al., 2021)**; I also worked with Professor Lucia Donatelli over hate speech detection in code-mixed Hindi-English, and with Professor Dipti Misra Sharma over a project in handling verb ellipsis in English-Hindi MT **(Bafna and Sharma, 2019)**.

Currently: I'm a Research Engineer at Inria, Paris, working on incorporating phoneme or morpheme level information into Transformer-based models, mentored by Benoît Sagot and Rachel Bawden. I'm investigating linguistically inspired architectural modifications of language models best suited to exploit both genealogical and contact-induced similarity between closely-related Indic languages (Cahyawijaya et al., 2021; Dhamecha et al., 2021), potentially including designing dynamic tokenizers, word-level task-based components (Nzeyimana and Niyongabo Rubungo, 2022), or input augmentation with linguistic information (in the absence of labelled data for the given dialects).

Future directions: My long-term research goal is to contribute to building theoretically well-motivated approaches to handling linguistic and dialectical diversity; my research interests present several aspects of this idea. In my opinion, a promising starting point of investigation is an exploration of modularization in Transformer-based models: that is, the attempt to directly or indirectly induced task or language based "components" of such models (Pfeiffer et al., 2022), in the context of language modelling for a family of closely-related languages/dialects. However, of course, this is only one of several interesting entry points to these questions. I also hope to use the insights obtained by such research to develop deployable and qualitatively tested language models for the low-resource North Indian dialect continuum, in continuation with my previous work with these languages. I aspire to participate in producing a body of work that makes fundamental progress on the ground level of language technology in the chosen line of research, as well as in theoretical and technical aspects.

Why a PhD?: As an undergraduate and Masters' student, I have tried to taste many different problems in NLP. These experiences helped me to streamline my research interests, as well as gain perspective on problems that excite me from various angles – from the work that goes into theoretically sound experiment design to the tricks (and patience!) one needs to work with less-than-perfect real world resources. A Ph.D. would be the ideal place for me to contribute towards problems that I find important and challenging, with the help of brilliant mentors and peers. I would love to pass on my love for NLP through teaching, as my teachers did for me; in the long term, I aim to be a professor and academic researcher.

Why JHU?: CLSP is especially attractive to me due to its exciting work in multilinguality and low-resource NLP as well as the number of faculty and researchers whose research interests are relevant to mine; examples of works I found particularly interesting are Lewis et al. (2020), Verma et al. (2022) and Tan et al. (2022). I would especially love to be supervised by Professors David Yarowsky, Philipp Koehn, or Kevin Duh, and I also believe that I have a strong research match with Dr. Mahsa Yarmohammadi. It would be a truly incredible opportunity for me to come to JHU.

References

Ayimunishagu Abulimiti and Tanja Schultz. 2020. Building language models for morphological rich low-resource languages using data from related donor languages: the case of uyghur. In *Proceedings of the 1st Joint Workshop on Spoken Language Technologies for Under-resourced languages (SLTU) and Collaboration and Computing for Under-Resourced Languages (CCURL)*, page 271–276, Marseille, France. European Language Resources association.

Mikel Artetxe, Gorka Labaka, and Eneko Agirre. 2017. Learning bilingual word embeddings with (almost) no bilingual data. In *Proceedings of the 55th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 451–462, Vancouver, Canada. Association for Computational Linguistics.

Samuel Cahyawijaya, Genta Indra Winata, Bryan Wilie, Karissa Vincentio, Xiaohong Li, Adhiguna Kuncoro, Sebastian Ruder, Zhi Yuan Lim, Syafri Bahar, Masayu Khodra, Ayu Purwarianti, and Pascale Fung. 2021. Indonlg: Benchmark and resources for evaluating indonesian natural language generation. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing*, page 8875–8898, Online and Punta Cana, Dominican Republic. Association for Computational Linguistics.

Yuan Chai, Yaobo Liang, and Nan Duan. 2022. Cross-lingual ability of multilingual masked language models: A study of language structure. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 4702–4712, Dublin, Ireland. Association for Computational Linguistics.

Tejas Indulal Dhamecha, Rudra Murthy V, Samarth Bharadwaj, Karthik Sankaranarayanan, and Pushpak Bhattacharyya. 2021. Role of language relatedness in multilingual fine-tuning of language models: A case study in indo-aryan languages. (arXiv:2109.10534). ArXiv:2109.10534 [cs].

Pratik Joshi, Sebastin Santy, Amar Budhiraja, Kalika Bali, and Monojit Choudhury. 2020. The state and fate of linguistic diversity and inclusion in the nlp world. In *Proceedings of the 58th Annual Meeting of the Association for Computational Linguistics*, page 6282–6293, Online. Association for Computational Linguistics.

Dylan Lewis, Winston Wu, Arya D. McCarthy, and David Yarowsky. 2020. Neural transduction for multilingual lexical translation. In *Proceedings of the 28th International Conference on Computational Linguistics*, page 4373–4384, Barcelona, Spain (Online). International Committee on Computational Linguistics.

Benjamin Minixhofer, Fabian Paischer, and Navid Rekabsaz. 2022. Wechsel: Effective initialization of subword embeddings for cross-lingual transfer of monolingual language models. In *Proceedings of the 2022 Conference of the North American Chapter of the Association for Computational Linguistics: Human Language Technologies*, page 3992–4006. ArXiv:2112.06598 [cs].

Antoine Nzeyimana and Andre Niyongabo Rubungo. 2022. Kinyabert: a morphology-aware kinyarwanda language model. In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, page 5347–5363, Dublin, Ireland. Association for Computational Linguistics.

Jonas Pfeiffer, Naman Goyal, Xi Victoria Lin, Xian Li, James Cross, Sebastian Riedel, and Mikel Artetxe. 2022. Lifting the curse of multilinguality by pre-training modular transformers. (arXiv:2205.06266). ArXiv:2205.06266 [cs].

Kartik Sharma, **Niyati Bafna**, and Samar Husain. 2021. Clause final verb prediction in hindi: Evidence for noisy channel model of communication. In *Proceedings of the Workshop on Cognitive Modeling and Computational Linguistics*, page 160–170, Online. Association for Computational Linguistics.

Jasdeep Singh, Bryan McCann, Richard Socher, and Caiming Xiong. 2019. Bert is not an interlingua and the bias of tokenization. In *Proceedings of the 2nd Workshop on Deep Learning Approaches for Low-Resource NLP (DeepLo 2019)*, page 47–55, Hong Kong, China. Association for Computational Linguistics.

Weiting Tan, Kevin Heffernan, Holger Schwenk, and Philipp Koehn. 2022. Multilingual representation distillation with contrastive learning. (arXiv:2210.05033). ArXiv:2210.05033 [cs].

Niyati Bafna and Dipti Sharma. 2019. Towards handling verb phrase ellipsis in english-hindi machine translation. In *Proceedings of the 16th International Conference on Natural Language Processing*, page 150–159, International Institute of Information Technology, Hyderabad, India. NLP Association of India.

Niyati Bafna, Martin Vastl, and Ondřej Bojar. 2021. Constrained decoding for technical term retention in english-hindi mt. In *Proceedings of the 18th International Conference on Natural Language Processing (ICON)*, page 1–6, National Institute of Technology Silchar, Silchar, India. NLP Association of India (NLPAI).

Niyati Bafna and Zdeněk Žabokrtský. 2022. Subword-based cross-lingual transfer of embeddings from hindi to marathi and nepali. In *Proceedings of the 19th SIGMORPHON Workshop on Computational Research in Phonetics, Phonology, and Morphology*, page 61–71, Seattle, Washington. Association for Computational Linguistics.

Neha Verma, Kenton Murray, and Kevin Duh. 2022. Strategies for adapting multilingual pretraining for domain-specific machine translation. In *Proceedings of the 15th biennial conference of the Association for Machine Translation in the Americas (Volume 1: Research Track)*, page 31–44, Orlando, USA. Association for Machine Translation in the Americas.

Yu Wan, Baosong Yang, Derek F. Wong, Lidia S. Chao, Haihua Du, and Ben C. H. Ao. 2020. Unsupervised neural dialect translation with commonality and diversity modeling. *Proceedings of the AAAI Conference on Artificial Intelligence*, 34(0505):9130–9137.

Winston Wu, Kevin Duh, and David Yarowsky. 2021. Sequence models for computational etymology of borrowings. In *Findings of the Association for Computational Linguistics: ACL-IJCNLP 2021*, page 4032–4037, Online. Association for Computational Linguistics.

Yi Zhu, Benjamin Heinzerling, Ivan Vulić, Michael Strube, Roi Reichart, and Anna Korhonen. 2019. On the importance of subword information for morphological tasks in truly low-resource languages. In *Proceedings of the 23rd Conference on Computational Natural Language Learning (CoNLL)*, page 216–226, Hong Kong, China. Association for Computational Linguistics.